Program : **Diploma in Cloud Computing and Big Data**	
Course Code : **3273**	Course Title: **Introduction to Cloud Computing**
Semester : **3**	Credits: **4**
Course Category: **Engineering Science**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To give a basic introduction to cloud computing.
- To give a basic idea about industrial cloud applications and services.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Computer Fundamentals	1008	Introduction to IT systems Lab	I
Programming concepts	2131	Problem Solving & Programming	II

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Outline fundamental concepts and models of cloud computing	15	Understanding
CO2	Summarize Principles of Parallel & Distributed Computing and virtualization	16	Understanding
CO3	Explain different Cloud Computing Mechanisms	15	Understanding
CO4	Contrast different public cloud platforms.	12	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						1
CO2	2						1
CO3	2						1
CO4	2						1

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Outline fundamental concepts and models of cloud computing**		
M1.01	Summarize basic concepts and Terminology in cloud computing	4	Understanding
M1.02	Extend Cloud Delivery Models	3	Understanding
M1.03	Classify Cloud Deployment Models	2	Understanding
M1.04	Explain Cloud Architecture	2	Understanding
M1.05	Summarize the history of Cloud Computing	2	Understanding
M1.06	Describe Cloud Computing Environments	2	Understanding

<table>
<tr><td colspan="4">Contents:
Basic Concepts and Terminology: Defining a cloud, IT Resource, Cloud Consumers and Cloud Providers, Scaling, Cloud Service, Cloud Service Consumer, Goals and Benefits, Risks and Challenges, Roles and Boundaries, Cloud Characteristics, Collaborating in Cloud Environment with example
Cloud Delivery Models- The cloud reference model - Infrastructure-as-a-service ((IaaS)), Platform-as-a-Service (PaaS), Software-as-a-Service (SaaS), Comparing Cloud Delivery Models, Combining Cloud Delivery Models
Cloud Deployment Models- Public Clouds, Community Clouds, Private Clouds, Hybrid Clouds, Other Cloud Deployment Models
Cloud Computing Architecture
History of Cloud Computing
Building Cloud Computing Environments: Application development, Infrastructure and system development, Computing Platform and Technologies.</td></tr>
<tr><td>CO2</td><td colspan="3">Summarize Principles of Parallel & Distributed Computing and Virtualization</td></tr>
<tr><td>M2.01</td><td>Compare Parallel and Distributed Computing</td><td>2</td><td>Understanding</td></tr>
<tr><td>M2.02</td><td>Explain Elements of Parallel Computing</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.03</td><td>Describe Elements of Distributed Computing</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.04</td><td>Explain the principle of Virtualization in cloud platform</td><td>6</td><td>Understanding</td></tr>
<tr><td>M2.05</td><td>Outline different Cloud Technology Examples</td><td>2</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test – I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:

Principles of Parallel and Distributed Computing: Parallel vs. Distributed computing, Elements of Parallel Computing- Hardware architectures for parallel processing, Approaches to parallel programming, Levels of parallelism
Elements of Distributed Computing- General concepts and definitions, Components of a distributed system
Virtualization: Introduction, Characteristics of virtualized environments, Taxonomy of virtualization techniques, Virtualization and Cloud Computing, Pros and cons of virtualization, Technology Examples</td></tr>
<tr><td>CO3</td><td colspan="3">Explain different Cloud Computing Mechanisms</td></tr>
<tr><td>M3.01</td><td>Explain Cloud Infrastructure Mechanisms</td><td>5</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Outline Specialized Cloud Mechanisms</td><td>6</td><td>Understanding</td></tr>
</table>

M3.03	Classify Cloud Management Mechanisms	4	Understanding

Contents :

Cloud Infrastructure Mechanisms: Logical Network Perimeter, Virtual Server, Cloud Storage Device, Cloud Usage Monitor, Resource Replication, Ready-Made Environment

Specialized Cloud Mechanisms: Automated Scaling Listener, Load Balancer, SLA Monitor, Pay-Per-Use Monitor, Audit Monitor, Failover System, Hypervisor, Resource Cluster, Multi-Device Broker, State Management Database.

Cloud Management Mechanisms: Remote Administration System, Resource Management System, SLA Management System, Billing Management System.

CO4	**Contrast different public cloud platforms.**		
M4.01	Explain Amazon web services	4	Understanding
M4.02	Describe Google App Engine	4	Understanding
M4.03	Outline Microsoft Azure	4	Understanding
	Series Test – II	1	

Contents :

Cloud Platforms in Industry : **Amazon web services** - Compute services - Storage services - Communication services.

Google AppEngine - Architecture and core concepts - Application life cycle, Cost model.

Microsoft Azure: Azure core concept, SQL Azure, Windows Azure platform appliance

Text / Reference

T/R	Book Title/Author
T1	Erl, Thomas, Ricardo Puttini, and Zaigham Mahmood. Cloud computing: concepts, technology, & architecture. Pearson Education, 2013.
T2	Rajkumar Buyya, Christian Vecchiola and Thamari Selvi S, —Mastering in Cloud Computing, McGraw Hill Education (India) Private Limited, 2013.

R1	Kai Hwang , Geoffrey C Fox, Jack J Dongarra : "Distributed and Cloud Computing – From Parallel Processing to the Internet of Things", Morgan Kaufmann Publishers – 2012.
R2	Dan C. Marinescu, Cloud Computing: Theory and Practice, Elsevier Science, 2013, 1st Edition, Print Book ISBN :9780124046276, eBook ISBN :9780124046412
R3	Rishabh Sharma, Cloud Computing Fundamentals, Industry Approaches and Trends, Wiley-2015

Online Resources

Sl.No	Website Link
1	https://csrc.nist.gov/publications/detail/sp/800-145/final
2	https://docs.aws.amazon.com/index.html?nc2=h_ql_doc#lang%2Fen_us
3	https://cloud.google.com/docs/
4	https://azure.microsoft.com/en-in/